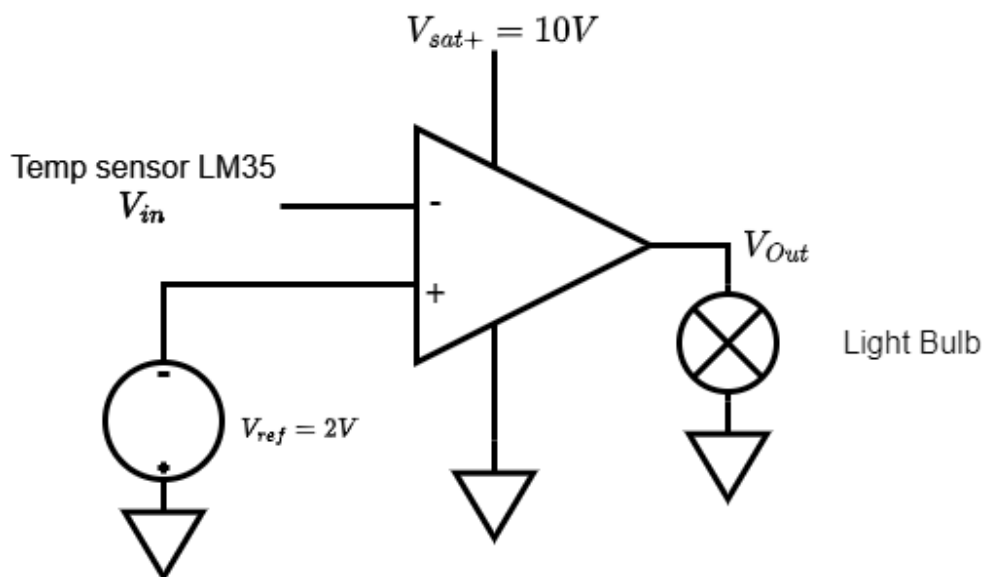


Sensor Question: (LM35 Temp. sensor)

1. Words: You have a temperature sensor (LM35). It gives a voltage of 1V at room temperature (25°C) and its voltage increase is 0.02V/°C. Design a circuit using Op-Amp so that a constant temperature of 30°C can be maintained in an incubator where heat is provided with a light bulb.

Ans: V_{ref} at 30°C = $1 + 0.2 \times 5 = 2V$ (25°C gives 1V & 0.2 V increases per degree)



2. Data Points: You have a temperature sensor (LM35). It gives the following voltage output at following temp

Temp(° C)	10	25	35
Voltage(V)	-2	1	3

Design a circuit using Op-Amp so that a constant temperature of 30°C can be maintained in an incubator where heat is provided with a light bulb.

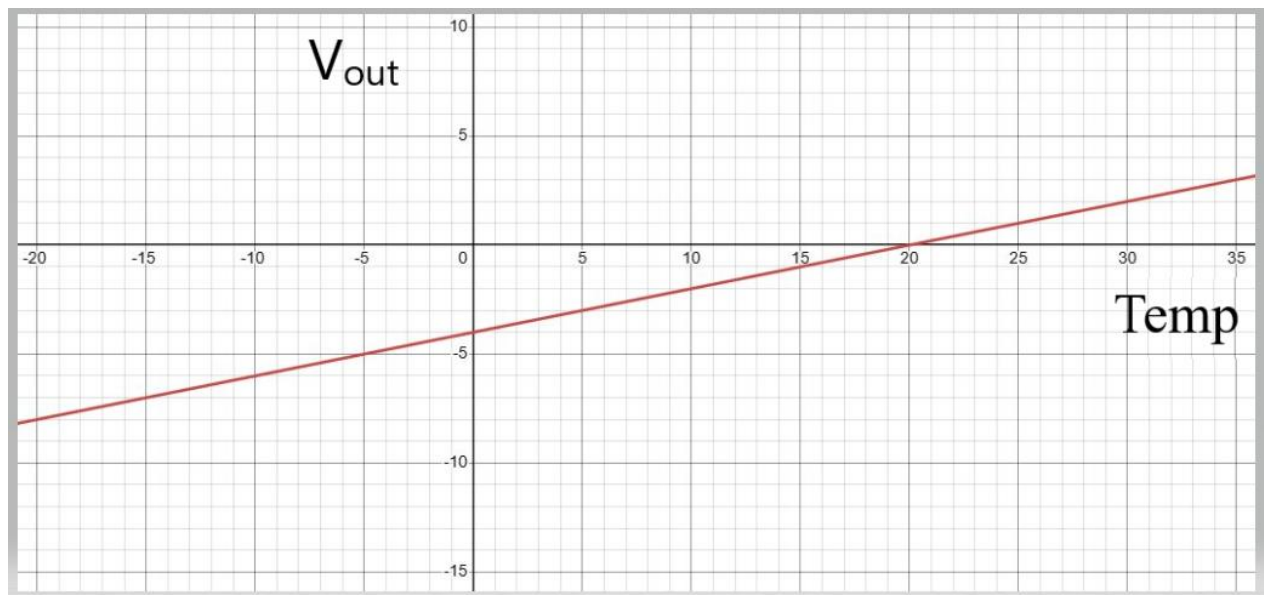
Ans: Find the function / straight line using the formula of finding the equation of a line using two points.

$$m(\text{slope}) = \frac{\text{change in } V}{\text{Change in } T} = \frac{-2 - 1}{10 - 25} = \frac{-3}{-15} = 0.2$$

$$V - V_1 = m(T - T_1) \equiv V - (-2) = 0.2(T - (-2)) \equiv V = 0.2T - 4$$

Then find V_{ref} at 30°C from $V = 0.2T - 4$ where $T=30$

3. Graph: You have a temperature sensor (LM35). It has a graph like the figure shown below. Design a circuit using Op-Amp so that a constant temperature of 30°C can be maintained in an incubator where heat is provided with a light bulb.



Ans. From graph find the slope, $m(0.2 \text{ here})$ then find the Y intercept which is $C(-4 \text{ here})$. Then you get your function $V(T)$. plug in the value of $T=30$ to the function to get the V_{ref} .

4. Function Given: You have a temperature sensor (LM35), It has a function $V(T) = 0.2T - 4$ where V is the output voltage, T input temp in Celsius. Design a circuit using Op-Amp so that a constant temperature of

30°C can be maintained in an incubator where heat is provided with a light bulb.

Ans: Given,

$$V(T) = 0.2T - 4$$

$$V(30) = 0.2 * 30 - 4 = 2$$

Rest is same